

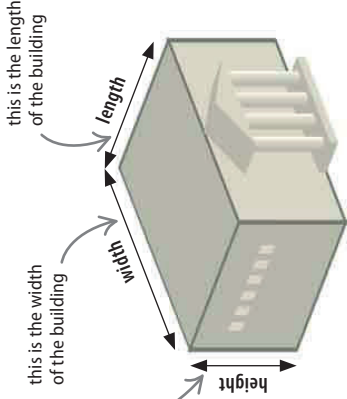
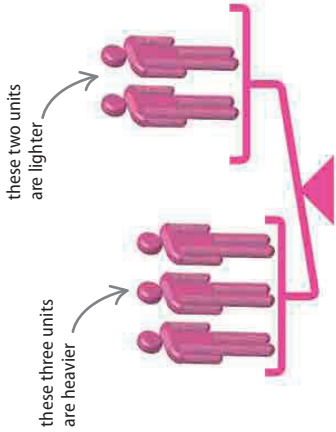


Units of measurement

UNITS OF MEASUREMENT ARE STANDARD SIZES USED TO MEASURE TIME, MASS, AND LENGTH.

Basic units

A unit is any agreed or standardized measurement of size. This allows quantities to be accurately measured. There are three basic units: time, weight (including mass), and length.



△ Time

Time is measured in milliseconds, seconds, minutes, hours, days, weeks, months, and years. Different countries and cultures may have calendars that start a new year at a different time.

△ Weight and mass

Weight is how heavy something is in relation to the force of gravity acting upon it. Mass is the amount of matter that makes up the object. Both are measured in the same units, such as grams and kilograms, or ounces and pounds.

△ Length

Length is how long something is. It is measured in centimeters, meters, and kilometers in the metric system, or in inches, feet, yards, and miles in the imperial system (see pp.242–245).

SEE ALSO

Volumes **154–155**

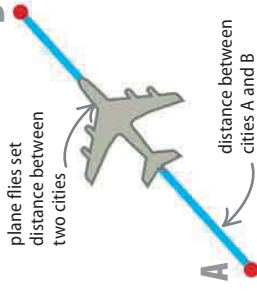
Formulas **177–179**

Reference **242–245**

LOOKING CLOSER

Distance

Distance is the amount of space between two points. It expresses length, but is also used to describe a journey, which is not always the most direct route between two points.

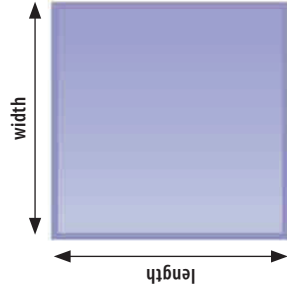


Compound measures

A compound unit is made up of more than one of the basic units, including using the same unit repeatedly. Examples include area, volume, speed, and density.

△ Area

Area is measured in squared units. The area of a rectangle is the product of its length and width; if they were both measured in meters (m) its area would be $m \times m$, which is written as m^2 .

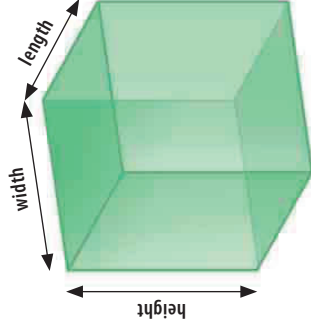


$$\text{area} = \text{length} \times \text{width}$$

area is made up of two of the same units, because width is also a length

△ Volume

Volume is measured in cubed units. The volume of a cuboid is the product of its height, width, and length; if they were all measured in meters (m), its area would be $m \times m \times m$, or m^3 .



$$\text{volume} = \text{length} \times \text{width} \times \text{height}$$

volume is a compound of three of the same units, because width and height are technically lengths